

Proposal

1. Project Description:

ClimaLens is a data analysis and visualization project designed to explore the causes and impacts of climate change on both the environment and human health. The project integrates data from global sources (**World Bank, WHO, OWID**), cleans it using Excel, and stores it in a structured SQL Server database. Through advanced queries, relationships, and analytical views, **ClimaLens** reveals connections between CO₂ emissions, water stress, and health indicators. The insights are visualized in Power BI, delivering an interactive dashboard that supports climate awareness and sustainable decision-making.

2. Group Members & Roles:

Omar Sayed: Responsible for coordinating tasks, managing the project timeline, ensuring deadlines are met, and overseeing communication among all team members.

Aly Yasser: Handles data extraction, cleaning, and transformation using Excel and SQL; designs and implements the relational database structure in SQL Server.

Ahmed Reda: Develops the Entity Relationship Diagram (ERD), defines primary and foreign keys, and ensures database normalization and referential integrity.

Doaa Ali: Builds and manages the ETL (Extract, Transform, Load) processes using SQL Server Integration Services (SSIS) to automate data import and integration from multiple sources into the Database

Malak Gamal: Writes analytical SQL queries, views, and stored procedures to extract insights and support the project's business questions.

Aya yasser & Omar Sayed: Creates and customizes interactive dashboards and visual reports to present key findings and trends effectively.

3.Team Leader:

Omar Sayed Darder Mohammed

4.Objectives:

1-To design and implement a relational database in SQL Server that organizes and integrates climate-related data from multiple trusted sources.

2-To perform data cleaning and preparation using Excel for accurate and consistent data storage.

3-To develop analytical SQL queries, views, and procedures that answer key business questions about climate change, environmental factors, and health outcomes.

4-To visualize insights and trends through interactive Power BI dashboards, enabling better understanding of the relationship between CO₂ emissions, water stress, and public health.

5-To support data-driven climate awareness and sustainable decision-making through integrated analysis and visualization tools.

5.Tools & Technologies:

Microsoft Excel: for data cleaning, preprocessing, and initial exploration.

SQL Server for database design, schema creation, data storage, and analytical querying.

SQL (DDL, DML, and DQL): for implementing views, joins, stored procedures, triggers, and aggregations.

Power BI: for interactive data visualization and dashboard development.

Lucid Chart: for creating the Entity Relationship Diagram (ERD) and database schema design.

GitHub: for version control and team collaboration.

6. Milestones & Deadlines:

Milestone	Description	Deadline
1. Data Collection & Source Review	Gather datasets from World Bank, WHO, and OWID; verify relevance and quality.	September 12, 2025
2. Data Cleaning & Preparation (Excel)	Clean, merge, and standardize raw datasets in Excel for database import.	September 20, 2025
3. Database Design (ERD & Schema)	Design logical and physical database structure using Draw.io and SQL Server.	September 30, 2025
4. Database Implementation	Create tables, relationships, constraints, and load sample data into SQL Server.	October 8, 2025
5. Query Development & Testing	Write and test SQL queries, joins, views, stored procedures, and triggers.	October 20, 2025
6. Data Analysis & Insight Generation	Perform analytical queries to answer business questions and extract insights.	October 31, 2025
7. Power BI Dashboard Development	Build and refine visual dashboards to present key findings and trends.	November 20, 2025

7.KPIs (Key Performance Indicators):

1. Data Cleaning & Processing

KPI: Percentage of missing or duplicate data handled during ETL and data preparation.

Target: 100%

2. Analysis & Insights

KPI: Percentage of agreed business questions answered through analysis.

Target: $\geq 90\%$

3. Visualization & Reporting

KPI 1: Average dashboard load time.

Target: < 3 seconds

KPI 2: Dashboard usability (percentage of users able to navigate without assistance).

Target: $\geq 80\%$

4. Final Documentation & Presentation

KPI 1: Final report completeness.

Target: 100%

KPI 2: Number of actionable recommendations included in the final report.

Target: ≥ 3